JSONata

Query and reshape JSON with compact expressions

Install & load

```
SET allow_extension_repositories = 'allowed';
CREATE EXTENSION REPOSITORY IF NOT EXISTS query_farm WITH PREFIX 'https://duckdb-extensions.query.farm';
INSTALL jsonata FROM query_farm;  LOAD jsonata FROM query_farm;
```

01 At a glance

Filter nested arrays, construct output objects, and aggregate values within a JSON document. JSONata keeps these transformations in one SQL expression and returns JSON that can be cast, extracted, or expanded by DuckDB.

JSON document → JSONata expression → New JSON shape

02 SQL entry point + expression syntax

`jsonata(expr, data)`

Evaluate an expression against a JSON document.

`jsonata(expr, data, bindings)`

Bind JSON object keys as $variables in the expression.

`Account.Name`

Follow nested fields using dotted paths.

`items[price > $limit]`

Filter array members by a predicate.

`{"total": $sum(prices)}`

Construct an output object with an aggregate.

`jsonata_validate(expr)`

Return whether an expression parses, without evaluating it.

`SET jsonata_timeout_ms / jsonata_max_depth`

Bound runtime (10 s) and recursion depth (1,000); raise them for heavy expressions.

03 Keep in mind

Bind values, keep expressions stable

Pass thresholds and other parameters through the bindings object instead of concatenating them into the expression. Its keys omit the dollar sign: {"limit": 100} is read as $limit inside JSONata.

04 Filter with a bound threshold

```
SELECT jsonata(
  '[items[price > $limit].id]',
  '{"items":[{"id":"A","price":20},' ||
  '{"id":"B","price":120}]}',
  '{"limit":100}'
);
-- ["B"]
```

The outer brackets preserve an array for one matching item.

05 Build a summary object

```
SELECT jsonata(
  '{"customer": name, "total": $sum(prices)}',
  '{"name":"Ada","prices":[12,18,5]}'
);
-- {"customer":"Ada","total":35}
```

Construct several output fields without unpacking the document.

06 Return a SQL number

```
SELECT jsonata('$sum(prices)',
  '{"prices":[12,18,5]}'
)::INTEGER AS total;
-- 35
```

Cast the JSON result before joining or doing typed SQL arithmetic.

07 In practice

Pay attention to result shape

JSONata sequences can collapse to a single value. Wrap a projection in array construction when downstream SQL expects JSON[]. The function returns JSON; cast numbers explicitly or use DuckDB JSON extractors for strings.

Regex uses RE2

Lookaround and backreferences raise invalid regular expression; the m flag is honoured. Expressions calling $now(), $millis() or $random() are evaluated for every row.

Full docs & examples ↗ Source ↗